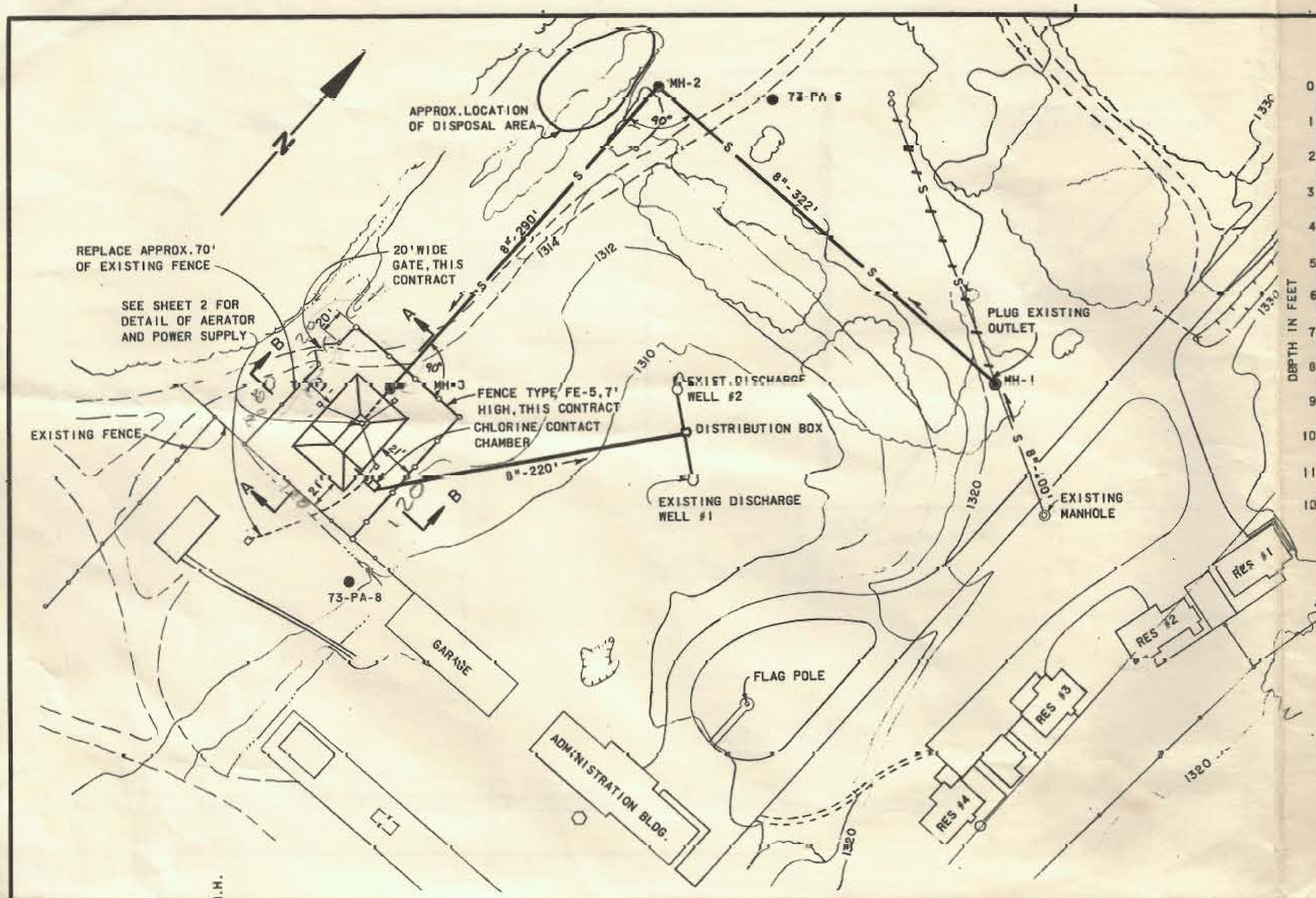
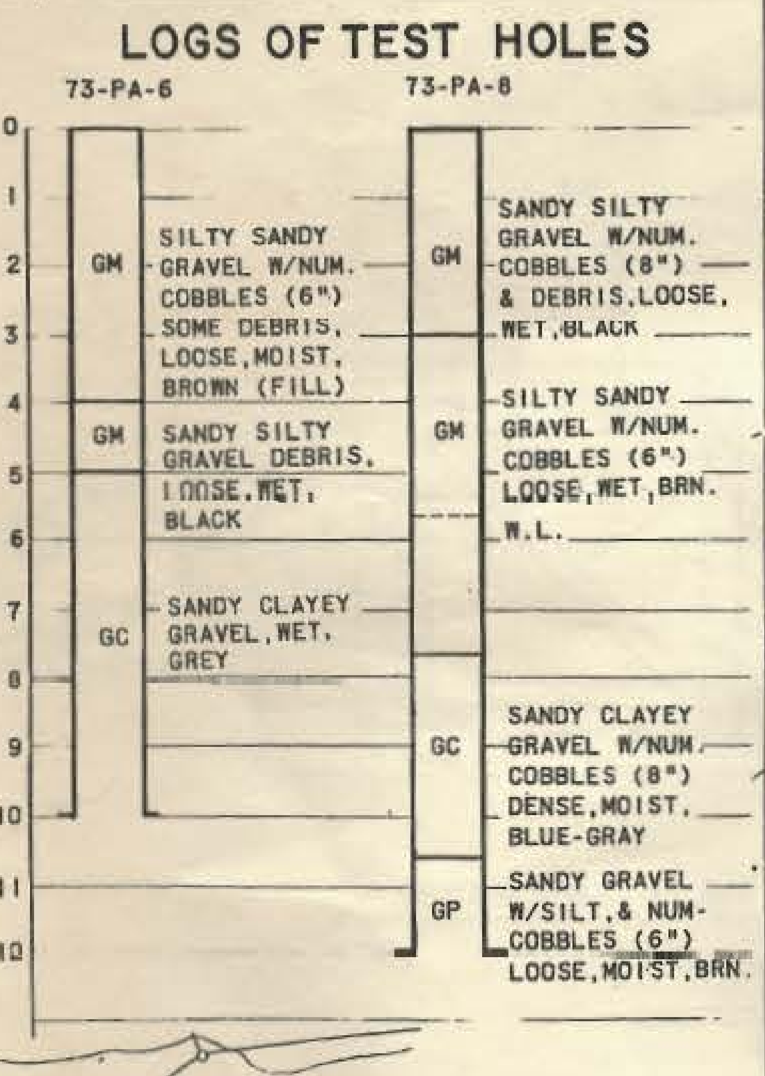

N
APPROX. LOCATION OF DISPOSAL AREA
MH-2
73-PA-6
REPLACE APPROX. 70' OF EXISTING FENCE
20' WIDE GATE, THIS CONTRACT
SEE SHEET 2 FOR DETAIL OF AERATOR AND POWER SUPPLY
8"-290'
8"-322'
PLUG EXISTING OUTLET
MH-3
FENCE TYPE FE-5, 7' HIGH, THIS CONTRACT
CHLORINE CONTACT CHAMBER
EXIST. DISCHARGE WELL #2
DISTRIBUTION BOX
MH-1
EXISTING FENCE
8"-220'
EXISTING DISCHARGE WELL #1
8"-100'
EXISTING MANHOLE
73-PA-8
GARAGE
FLAG POLE
ADMINISTRATION BLDG.
RES #1
RES #2
RES #3
RES #4
1310
1312
1314
1320
1330

PLAN

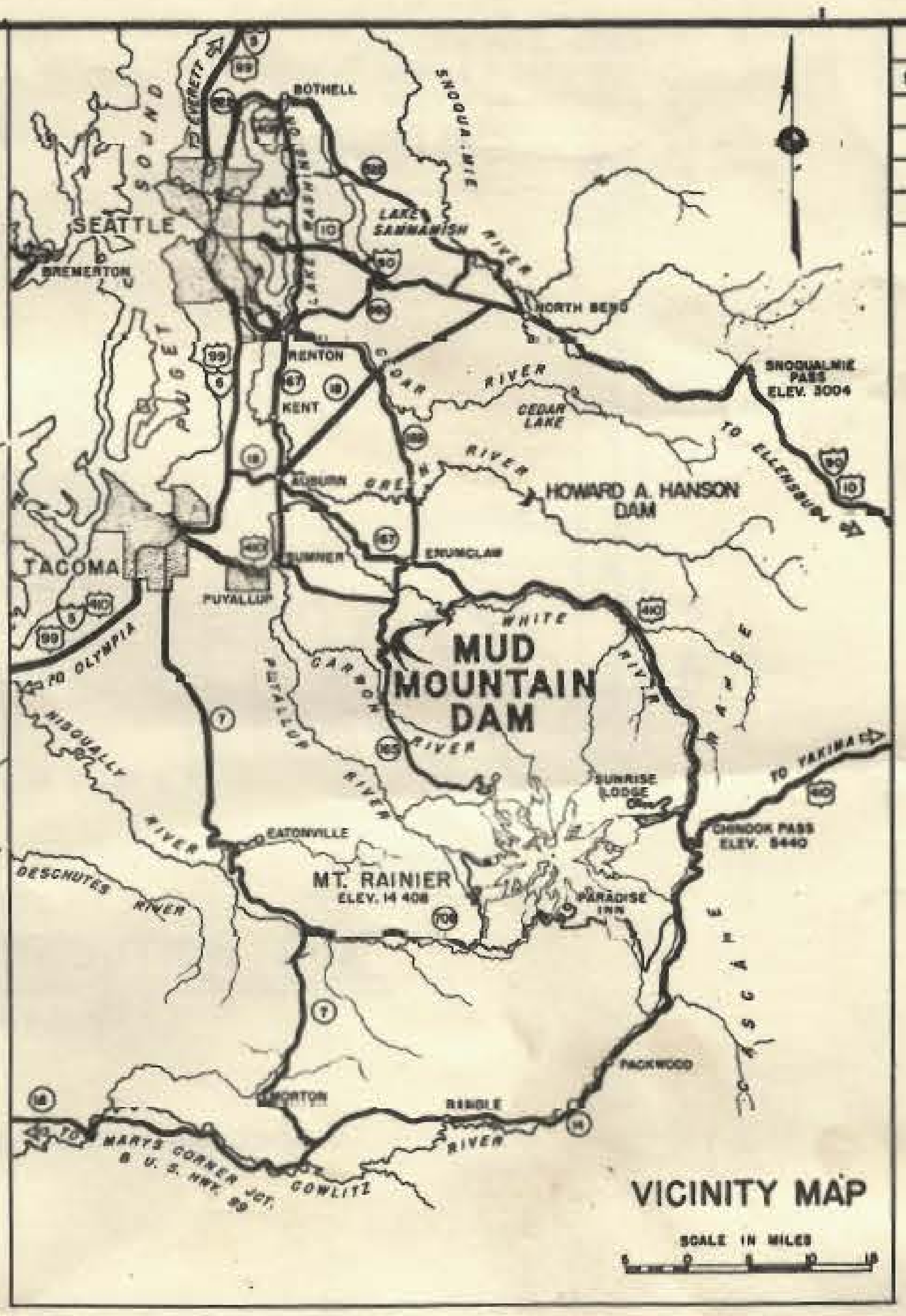
SCALE: 1" = 50'

LOGS OF TEST HOLES



73-PA-6	
GM	SILTY SANDY GRAVEL W/NUM. COBBLES (6") SOME DEBRIS, LOOSE, MOIST, BROWN (FILL)
GM	SANDY SILTY GRAVEL DEBRIS, LOOSE, WET, BLACK
GC	SANDY CLAYEY GRAVEL, WET, GREY

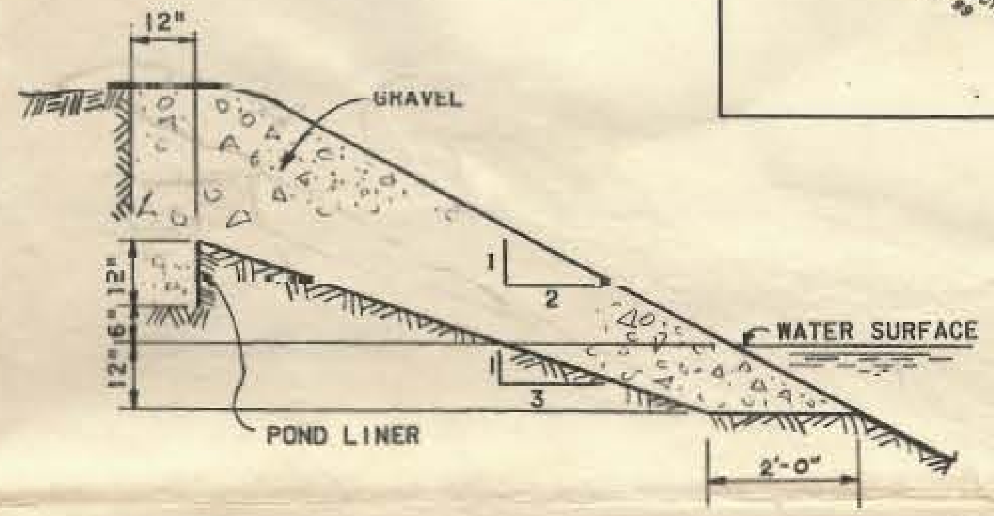
73-PA-8	
GM	SANDY SILTY GRAVEL W/NUM. COBBLES (8") & DEBRIS, LOOSE, WET, BLACK
GM	SILTY SANDY GRAVEL W/NUM. COBBLES (6") LOOSE, WET, BRN. W.L.
GC	SANDY CLAYEY GRAVEL W/NUM. COBBLES (8") DENSE, MOIST, BLUE-GRAY
GP	SANDY GRAVEL W/SILT, & NUM- COBBLES (6") LOOSE, MOIST, BRN.


SEATTLE
BOTHELL
BREMERTON
LAKE SAMMAMISH
RENTON
KENT
NORTH BEND
SNOQUALMIE PASS ELEV. 3004
CEDAR LAKE
TO ELLENSBURG
AUBURN
HOWARD A. HANSON DAM
TACOMA
SUMNER
ENUMCLAW
TO OLYMPIA
PUYALLUP
WHITE RIVER
MUD MOUNTAIN DAM
NISQUALLY RIVER
CARBON RIVER
PUYALLUP RIVER
TO YAKIMA
EATONVILLE
SUNRISE LODGE
CHINOOK PASS ELEV. 5440
DESCHUTES RIVER
MT. RAINIER ELEV. 14 408
PARADISE INN
MORTON
PACKWOOD
RANDLE
COWLITZ RIVER
MARYS CORNER JCT. U.S. HWY 99
SCALE IN MILES

VICINITY MAP

REVISIONS

SYMBOL	ZONE	DESCRIPTION	DATE	BY

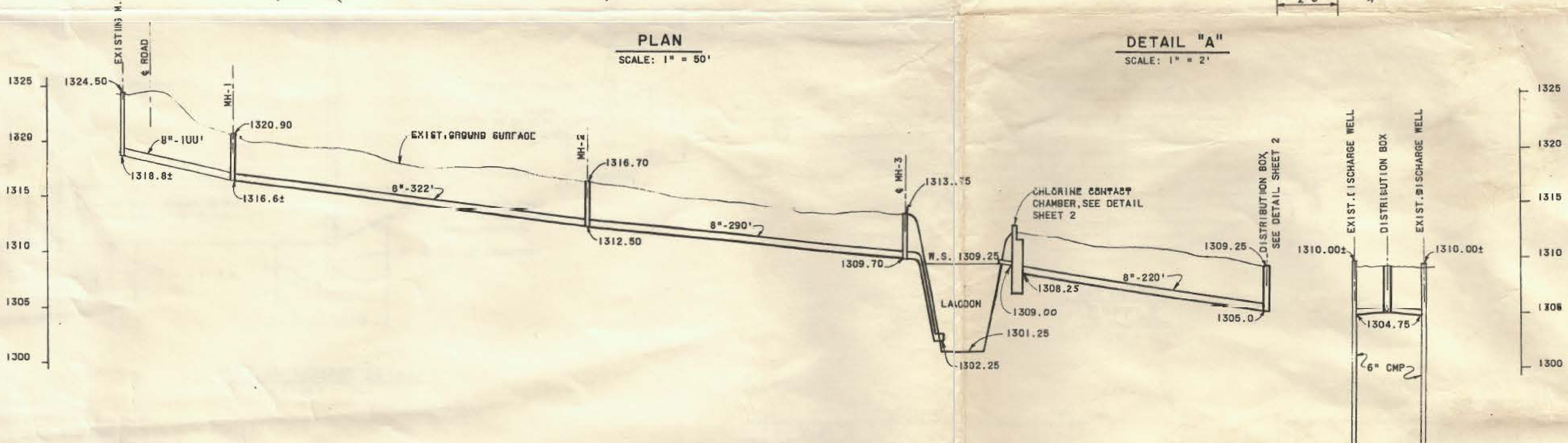

12"
GRAVEL
12"
6"
12"
1
2
1
3
WATER SURFACE
POND LINER
2'-0"

DETAIL "A"

SCALE: 1" = 2'

LEGEND

EXISTING OR BY OTHERS		THIS CONTRACT
	SECURITY FENCE	
◎	MANHOLE	◉
	PAVED ROAD	
	GRAVEL ROAD	
1325	GROUND SURFACE CONTOUR	
8"-100' S	SEWER LINE, SIZE & LENGTH	8"-100' S
	SEWER TO BE REMOVED	
▭	BUILDING OR STRUCTURE	
73-PA-8 ●	POWER-AUGER TEST HOLE & NO.	
CMP	CORRIGATED METAL PIPE	

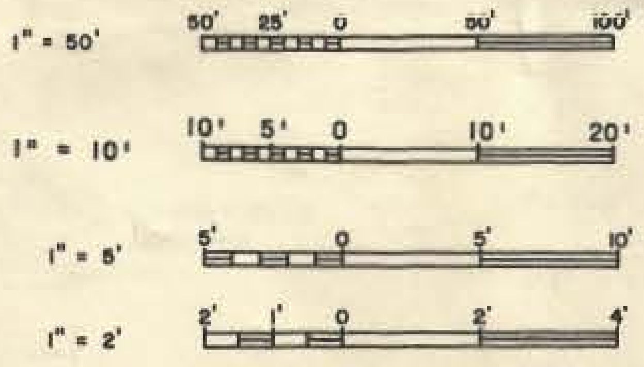

EXISTING M.H.
℄ ROAD
1325
1320
1315
1310
1305
1300
1324.50
8"-100'
1318.8±
MH-1
1320.90
1316.6±
EXIST. GROUND SURFACE
8"-322'
MH-2
1316.70
1312.50
8"-290'
℄ MH-3
1313.75
1309.70
W.S. 1309.25
LAGOON
1302.25
1301.25
1309.00
1308.25
CHLORINE CONTACT CHAMBER, SEE DETAIL SHEET 2
8"-220'
1309.25
1305.0
DISTRIBUTION BOX, SEE DETAIL SHEET 2
EXIST. DISCHARGE WELL
DISTRIBUTION BOX
EXIST. DISCHARGE WELL
1310.00±
1310.00±
1304.75
6" CMP

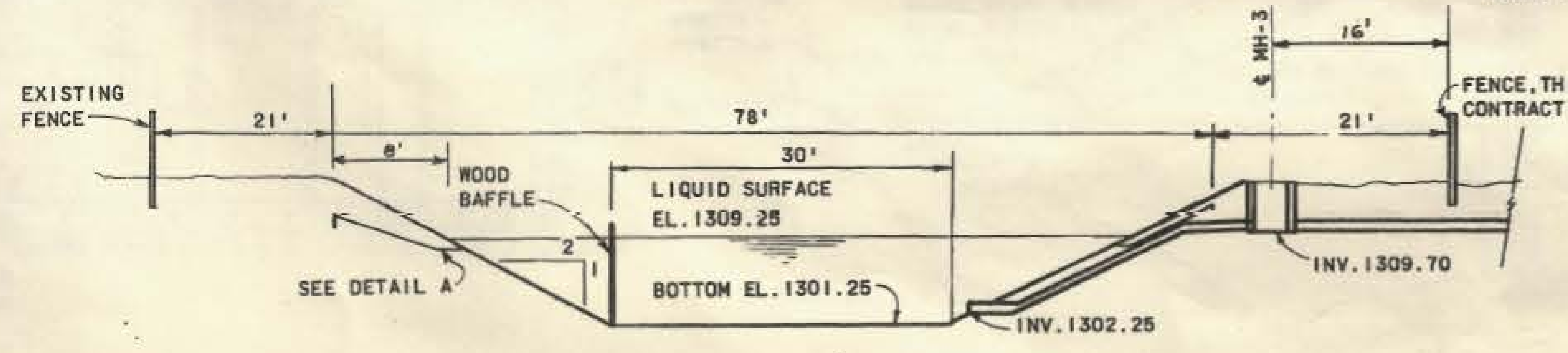
STANDARD DRAWINGS BOUND IN SPECIFICATIONS

DWG	SHEET	TITLE
495/40-16-16	1	FENCE AND DETAILS
495/71-07-02		SHALLOW SANITARY SEWER MANHOLE

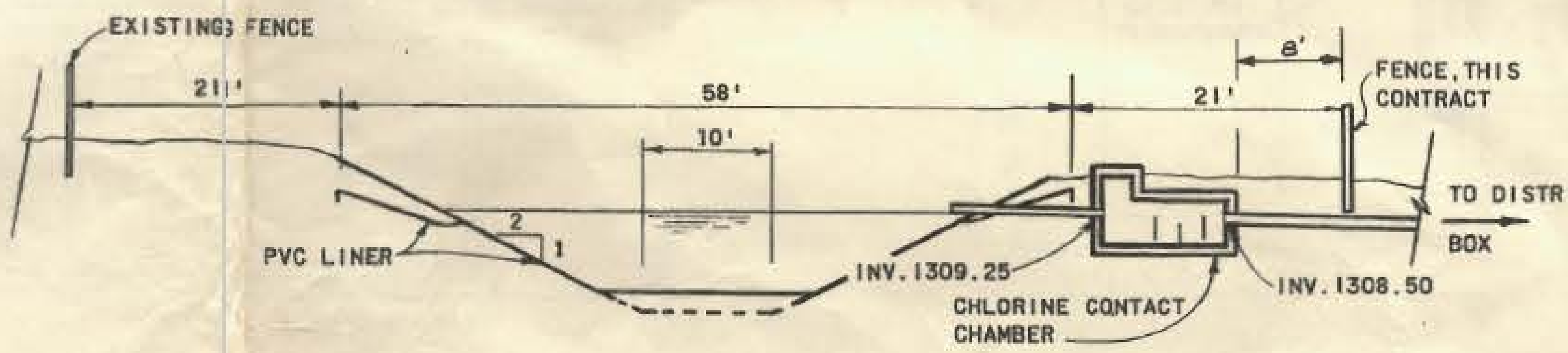
PROFILE

SCALE: HORIZ. 1" = 50'
VERT. 1" = 5'


1" = 50'
50'
25'
0
50'
100'
1" = 10'
10'
5'
0
10'
20'
1" = 5'
5'
0
5'
10'
1" = 2'
2'
1'
0
2'
4'


EXISTING FENCE
21'
8'
78'
30'
℄ MH-3
16'
21'
FENCE, THIS CONTRACT
WOOD BAFFLE
LIQUID SURFACE EL. 1309.25
2
1
SEE DETAIL A
BOTTOM EL. 1301.25
INV. 1309.70
INV. 1302.25

SECTION A-A


EXISTING FENCE
21'
58'
10'
8'
21'
FENCE, THIS CONTRACT
TO DISTRIBUTION BOX
PVC LINER
2
1
INV. 1309.25
CHLORINE CONTACT CHAMBER
INV. 1308.50

SECTION B-B

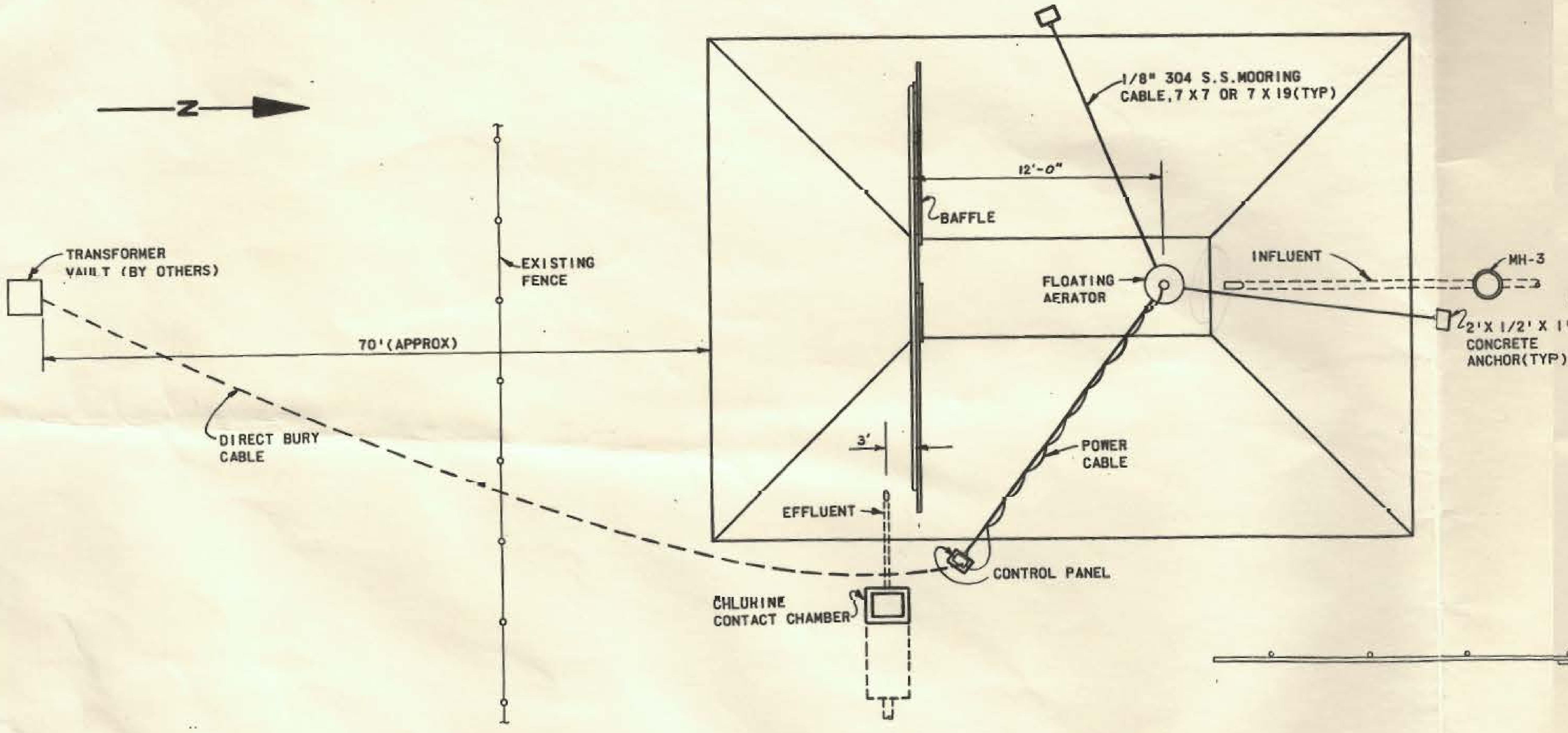
SCALE: 1" = 10'

THIS DRAWING HAS BEEN REDUCED TO APPROXIMATELY ONE-HALF THE ORIGINAL SCALE.

SUBMITTED:	U. S. ARMY ENGINEER DISTRICT, SEATTLE CORPS OF ENGINEERS SEATTLE, WASHINGTON
CHIEF, DESIGN BRANCH	SEWAGE TREATMENT FACILITY PLAN PROFILE AND SECTIONS MUD MOUNTAIN DAM
RECOMMENDED: Fred H. Oebe CHIEF, ENGINEERING DIVISION	WHITE RIVER — WASHINGTON
APPROVED: Frederick W. Mueller For COL. C.O.E. DISTRICT ENGINEER	SIZE: F — INVITATION NO. DACW 67-74 B-0056 — FILE NO. E-3-8-278 — PLATE
DATE: 1974 FEB 19	DSGN. METTLER — CHK. BYERS — SHEET 1

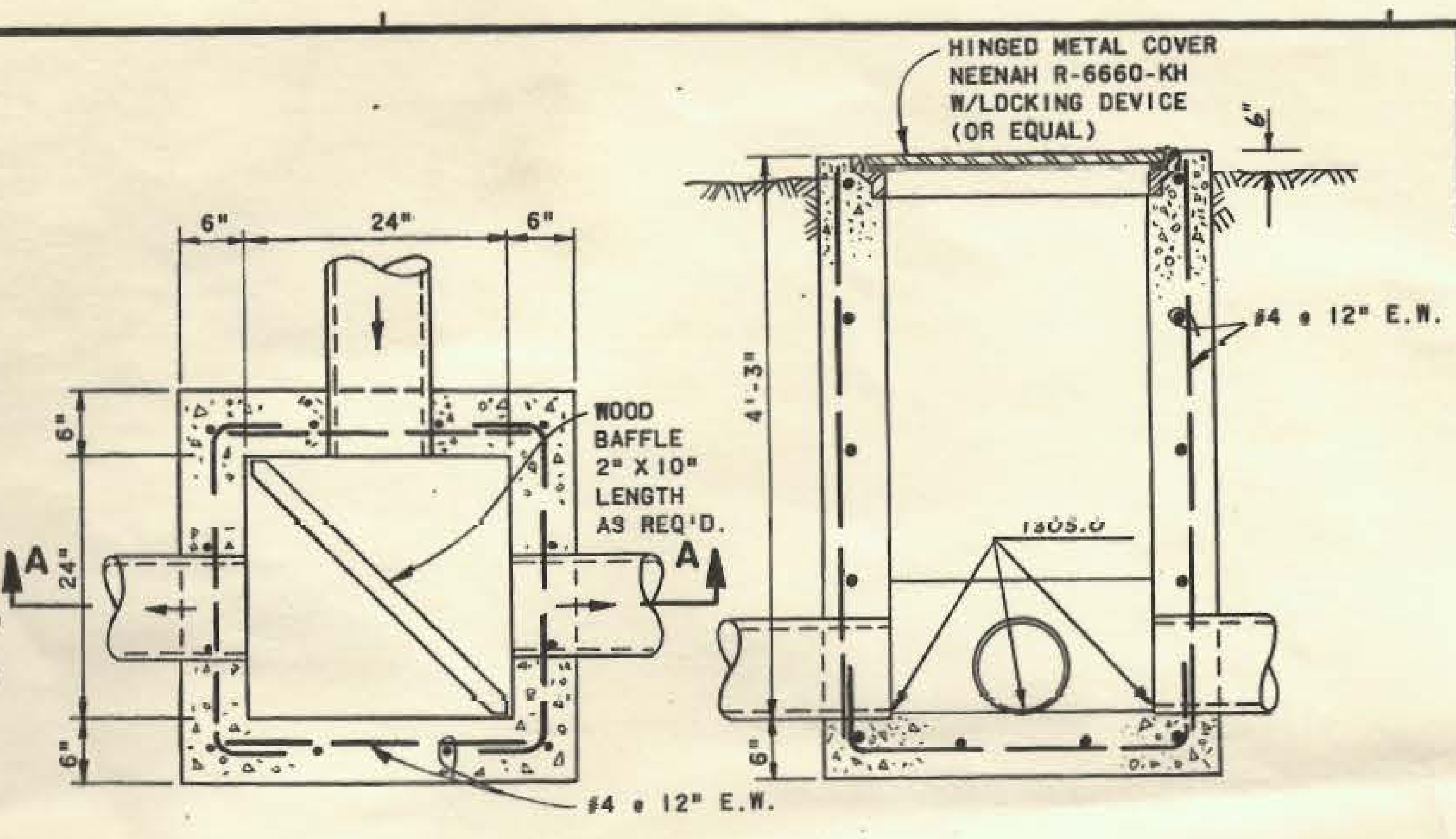
SYMBOL	ZONE	DESCRIPTION	DATE	BY

REVISIONS


N
TRANSFORMER VAULT (BY OTHERS)
EXISTING FENCE
70'(APPROX)
DIRECT BURY CABLE
1/8" 304 S.S.MOORING CABLE,7 X 7 OR 7 X 19(TYP)
12'-0"
BAFFLE
FLOATING AERATOR
INFLUENT
MH-3
2'X 1/2'X 1' CONCRETE ANCHOR(TYP)
3'
POWER CABLE
EFFLUENT
CONTROL PANEL
CHLORINE CONTACT CHAMBER

LAGOON PLAN

SCALE: 1/8" = 1'-0"

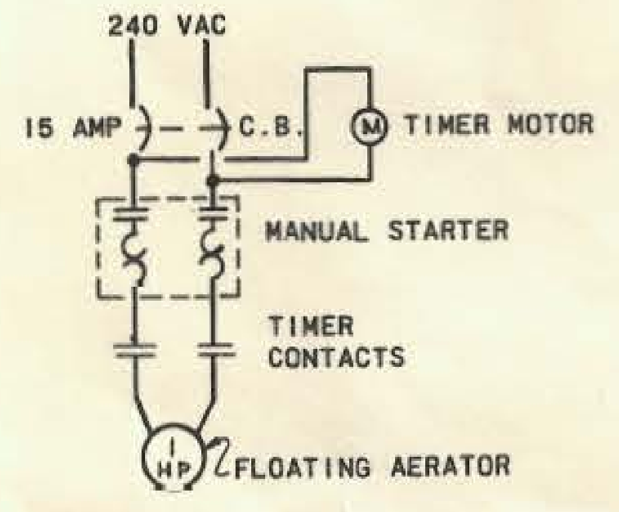

HINGED METAL COVER NEENAH R-6660-KH W/LOCKING DEVICE (OR EQUAL)
6"
6"
24"
6"
WOOD BAFFLE 2" X 10" LENGTH AS REQ'D.
A
A
24"
4'-3"
1305.0
#4 @ 12" E.W.
#4 @ 12" E.W.

SECTIONAL PLAN

SECTION A-A

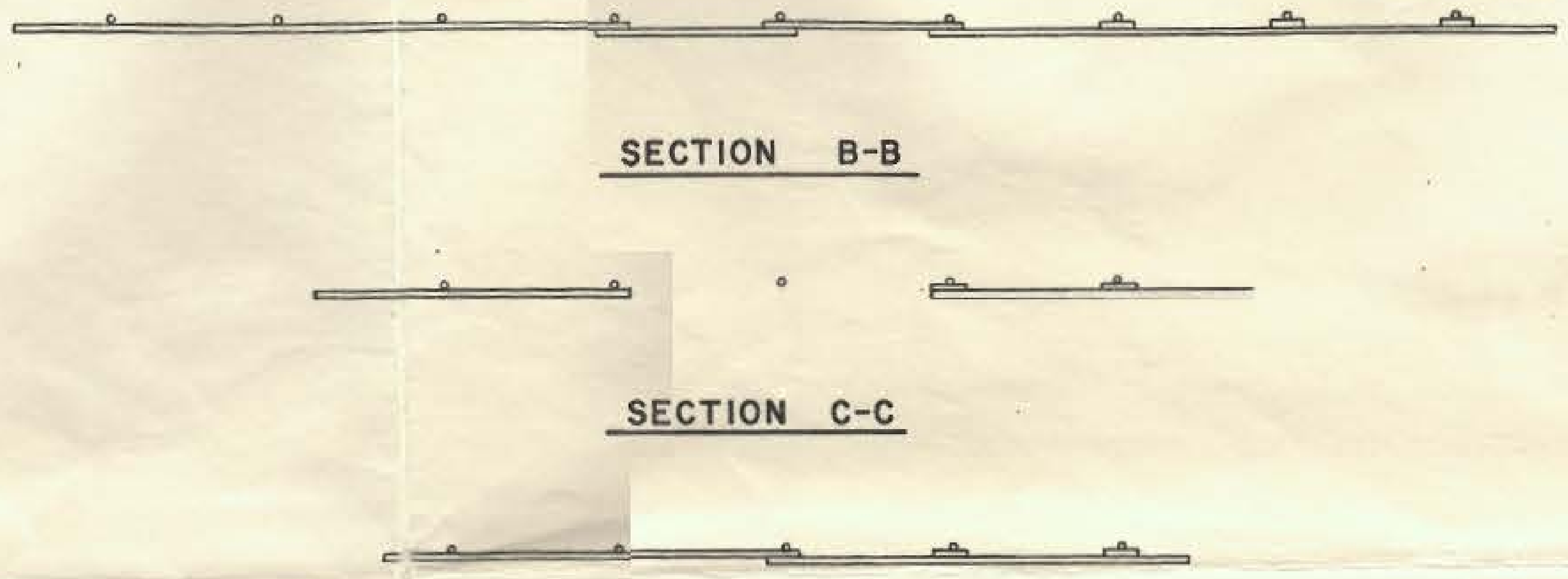
DISTRIBUTION BOX

SCALE: 1" = 1'-0"


240 VAC
15 AMP
C.B.
M
TIMER MOTOR
MANUAL STARTER
TIMER CONTACTS
HP
FLOATING AERATOR

FLOATING AERATOR CONTROL DIAGRAM

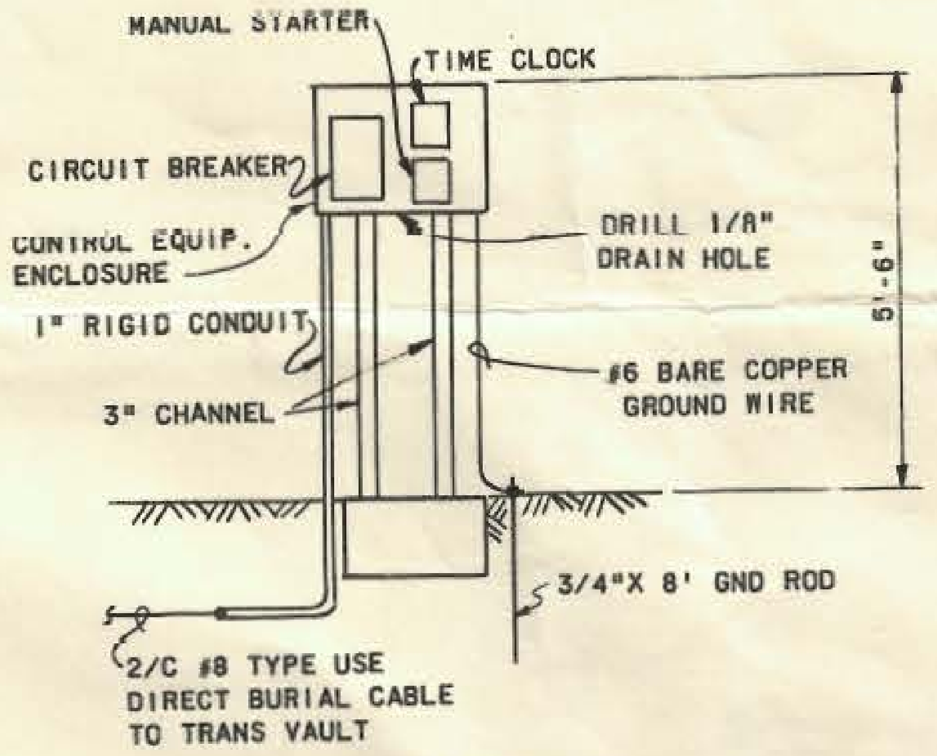
NOT TO SCALE



SECTION B-B

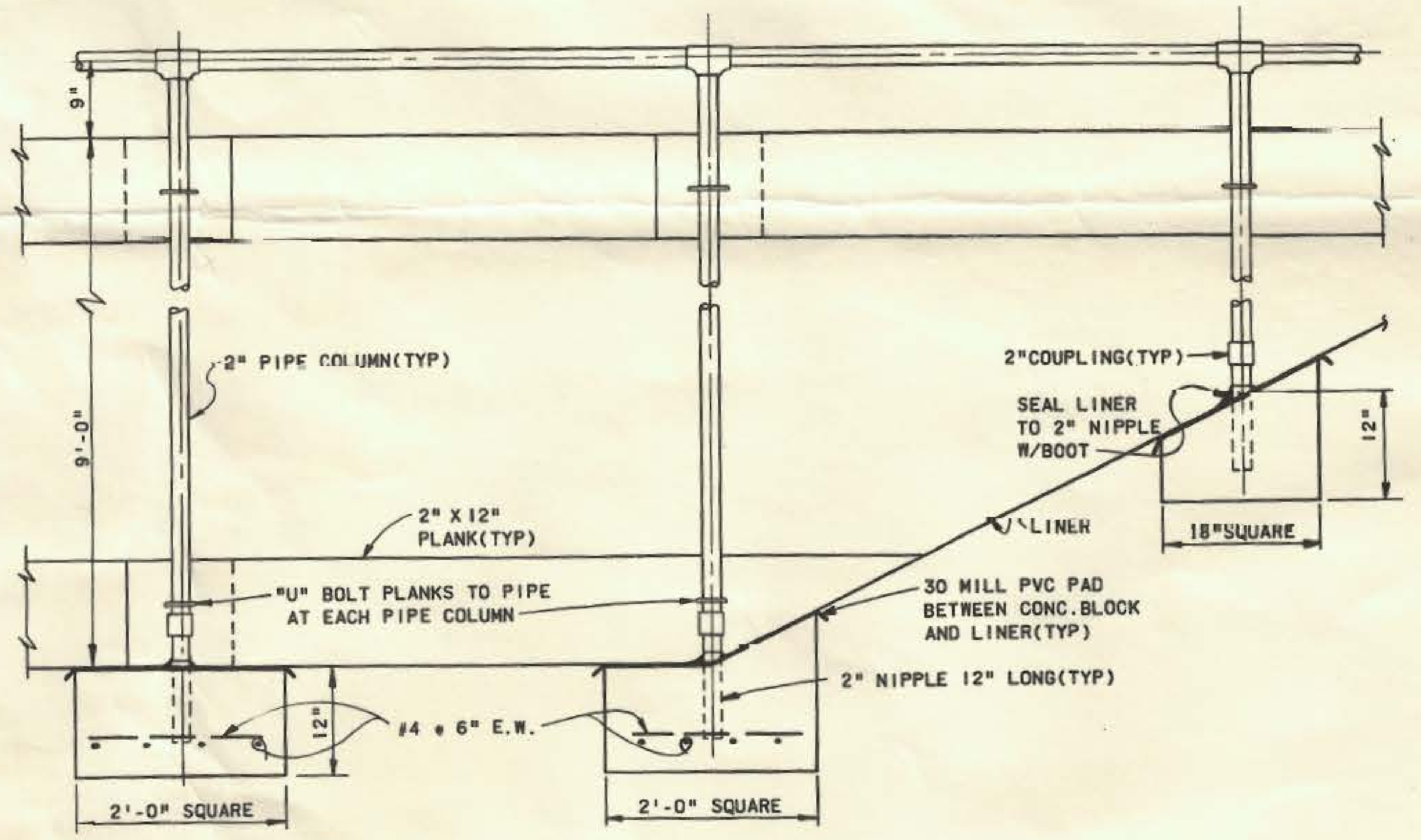
SECTION C-C

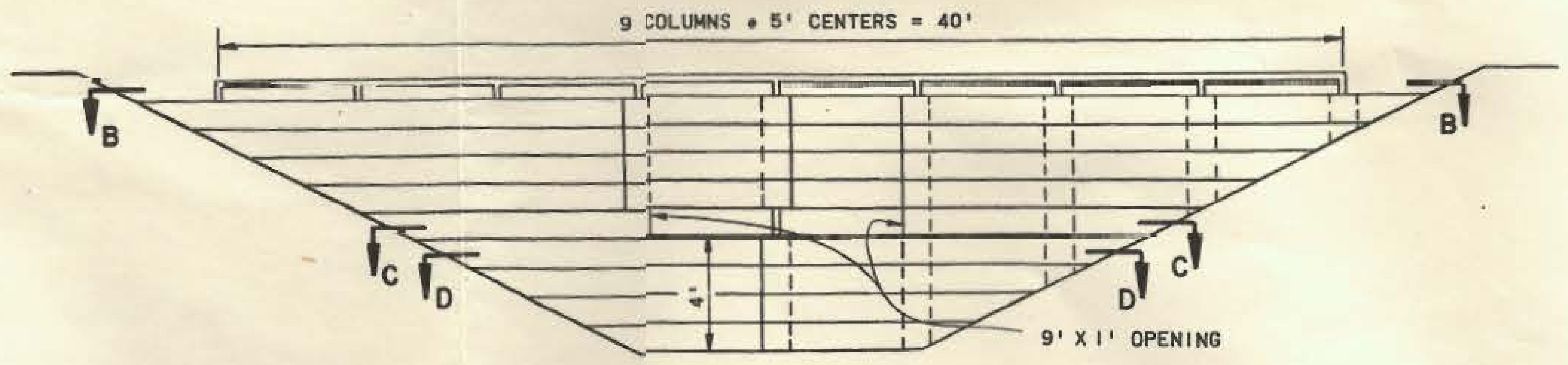
SECTION D-D


MANUAL STARTER
TIME CLOCK
CIRCUIT BREAKER
CONTROL EQUIP. ENCLOSURE
DRILL 1/8" DRAIN HOLE
1" RIGID CONDUIT
5'-6"
#6 BARE COPPER GROUND WIRE
3" CHANNEL
3/4"X 8' GND ROD
2/C #8 TYPE USE DIRECT BURIAL CABLE TO TRANS VAULT

CONTROL PANEL DETAIL

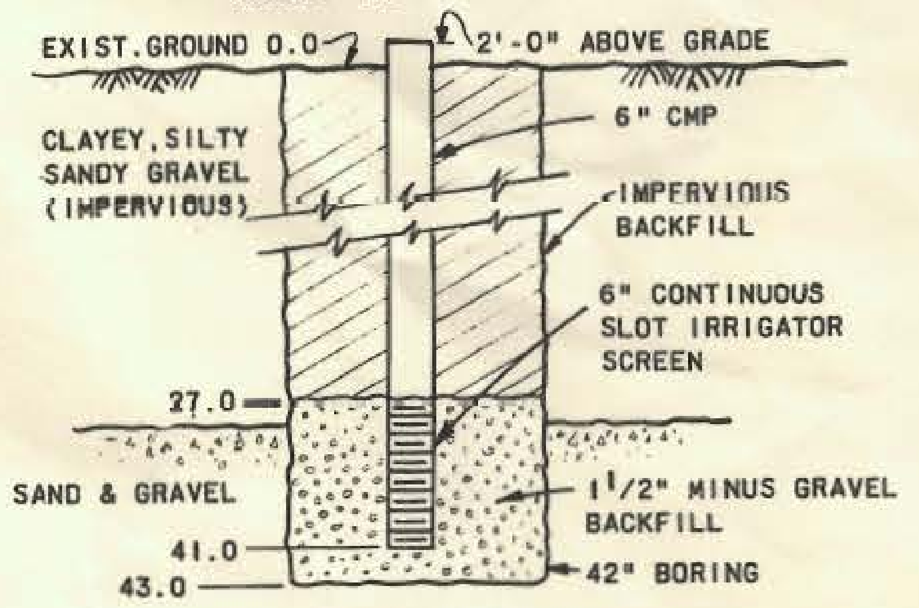
SCALE: 1/2" = 1'-0"


9"
2" PIPE COLUMN(TYP)
9'-0"
2"COUPLING(TYP)
SEAL LINER TO 2" NIPPLE W/BOOT
12"
18"SQUARE
LINER
2" X 12" PLANK(TYP)
"U" BOLT PLANKS TO PIPE AT EACH PIPE COLUMN
30 MILL PVC PAD BETWEEN CONC.BLOCK AND LINER(TYP)
2" NIPPLE 12" LONG(TYP)
12"
#4 @ 6" E.W.
2'-0" SQUARE
2'-0" SQUARE


9 COLUMNS @ 5' CENTERS = 40'
B
B
C
D
D
C
4'
9' X 1' OPENING

ELEVATION BAFFLE

SCALE: 1/4" = 1'-0"

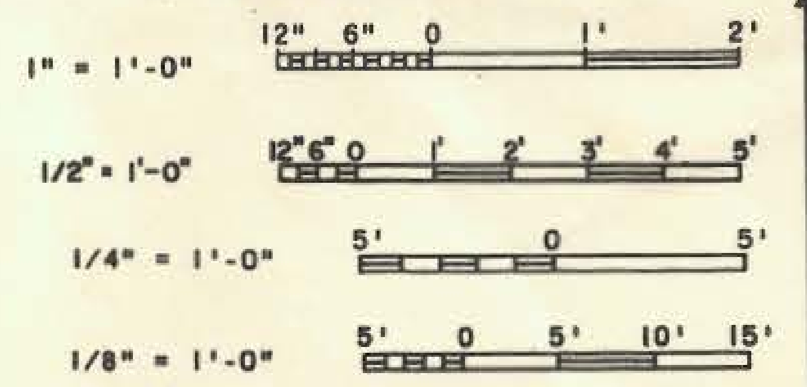

EXIST.GROUND 0.0
2'-0" ABOVE GRADE
6" CMP
CLAYEY,SILTY SANDY GRAVEL (IMPERVIOUS)
IMPERVIOUS BACKFILL
6" CONTINUOUS SLOT IRRIGATOR SCREEN
27.0
SAND & GRAVEL
1 1/2" MINUS GRAVEL BACKFILL
41.0
43.0
42" BORING

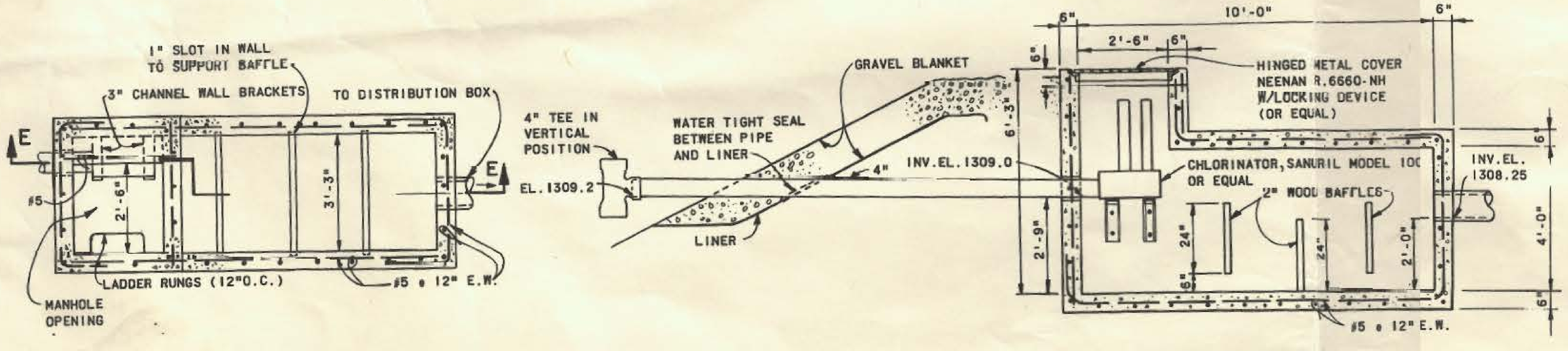
PIPE ANCHORAGE DETAIL

SCALE: 1" = 1'-0"

EXISTING DISCHARGE WELLS 1 & 2

NOT TO SCALE


1" = 1'-0"
12" 6" 0 1' 2'
1/2" = 1'-0"
12" 6" 0 1' 2' 3' 4' 5'
1/4" = 1'-0"
5' 0 5'
1/8" = 1'-0"
5' 0 5' 10' 15'


1" SLOT IN WALL TO SUPPORT BAFFLE
3" CHANNEL WALL BRACKETS
TO DISTRIBUTION BOX
E
E
#5
2'-6"
3'-3"
LADDER RUNGS (12"O.C.)
#5 @ 12" E.W.
MANHOLE OPENING
4" TEE IN VERTICAL POSITION
EL. 1309.2
WATER TIGHT SEAL BETWEEN PIPE AND LINER
GRAVEL BLANKET
4"
LINER
6"
10'-0"
6"
2'-6"
6"
6"
HINGED METAL COVER NEENAH R.6660-NH W/LOCKING DEVICE (OR EQUAL)
6'-3"
INV.EL. 1309.0
CHLORINATOR, SANURIL MODEL 100 OR EQUAL
2" WOOD BAFFLES
INV.EL. 1308.25
6"
4'-0"
2'-9"
24"
24"
2'-0"
6"
#5 @ 12" E.W.

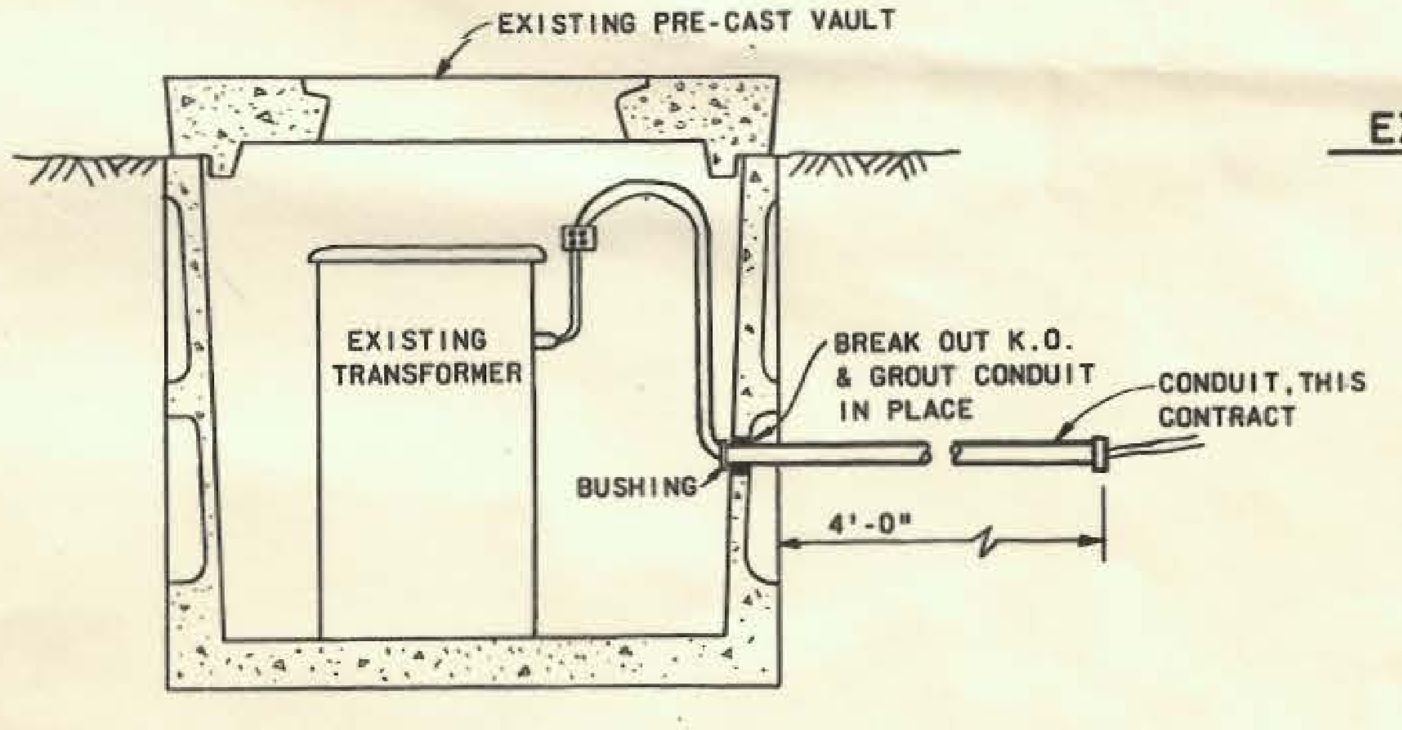
PLAN

SCALE: 1/2" = 1'-0"

SECTION E-E

SCALE: 1/2" = 1'-0"

CHLORINATOR CHAMBER


EXISTING PRE-CAST VAULT
EXISTING TRANSFORMER
BREAK OUT K.O. & GROUT CONDUIT IN PLACE
CONDUIT, THIS CONTRACT
BUSHING
4'-0"

DETAIL AT ELEC. VAULT

SCALE: 1" = 1'-0"

THIS DRAWING HAS BEEN REDUCED TO APPROXIMATELY ONE-HALF THE ORIGINAL SCALE.

PREPARED: James D. Cernisse
CHIEF, ELECT-MECH SECTION
SUBMITTED: Edwin Durrah
CHIEF, DESIGN BRANCH
DATE: 1974 FEB 19

U. S. ARMY ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON

SEWAGE TREATMENT FACILITY
DETAILS
MUD MOUNTAIN DAM
WHITE RIVER WASHINGTON

SIZE	INVITATION NO.	FILE NO.	PLATE
F		E-3-8-278	

DSGN. METTLER
CHK. BYERS
SHEET 2

SANURIL®

WASTEWATER CHLORINATOR

For small treatment plants up to 10,000 GPD*

MINI-SAN™

MODEL

100

THE SIMPLE, RELIABLE WAY TO DISINFECT WASTEWATER

The SANURIL *Mini-San* Model 100 Wastewater Chlorinator is a complete one-piece system for highly reliable disinfection of treated effluent from any sewage plant having a design capacity of 10,000 GPD or less. The chlorinator is used in conjunction with SANURIL 115 Tablets, a new disinfecting agent that provides a chemically stable source of chlorine for wastewater disinfection.

The SANURIL *Mini-San* Chlorinator can be installed in existing treatment plants where wastewater chlorination is being added or *to replace troublesome, unreliable or complicated chlorination systems.* It is equally suitable for factory assembly into new package plants. Typical installations have included a wide range of sites such as:

- Restaurants
- Service Stations
- Highway Rest Stops
- Drilling Platforms
- Trailer Parks
- Small Industrial Plants
- Recreation Parks
- Camping Facilities
- Schools and many others

SANURIL — SUPERIOR PERFORMANCE AT A LOWER COST

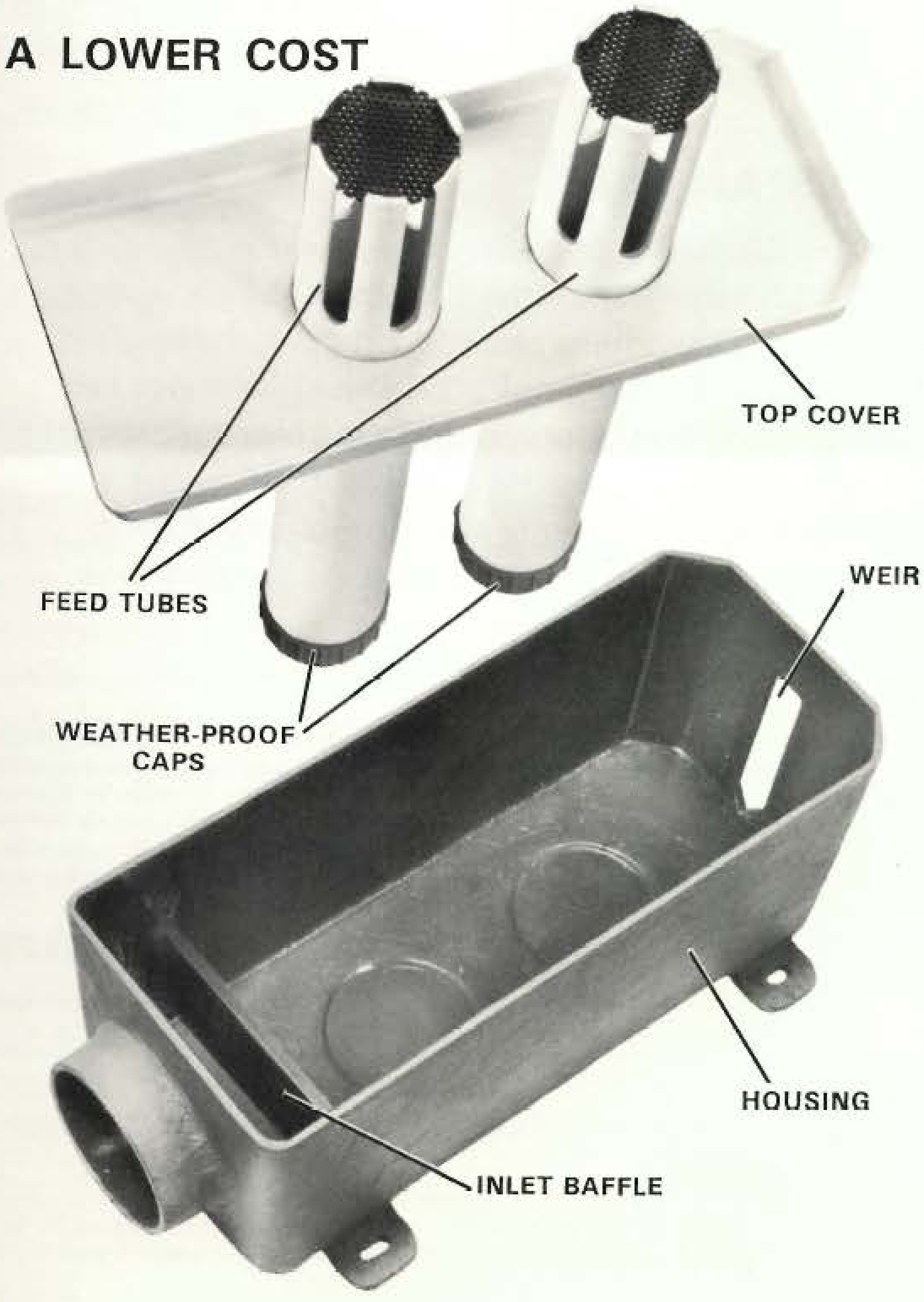
Better Performance — The entire plant flow passes through the chlorinator where thorough mixing of wastewater with a controlled amount of chlorine takes place. This means more effective bacteriacidal action and less bacteria regrowth.

Low Initial Cost — In addition to low investment in the *Mini-San* Chlorinator, savings are compounded because no pumps, mixing tanks or costly control devices are required as with most other chlorination systems.

Simple, Low-Cost Installation — One-piece chlorinator construction requires only inlet piping. No electrical power, wiring, bypasses or other piping required.

Lower Operating Costs — Chlorinator operates up to 60 days on a single charge. No time-consuming pre-mixing of chemicals or costly cylinder handling. Recharging SANURIL 115 tablets is quick and easy. No costs for electrical power or auxiliary power.

Maintenance-Free Operation — No tanks, pumps, cylinders, external control devices or other moving parts to break down. System is non-mechanical and rust and corrosion proof.



**For plants over 10,000 GPD flow use the SANURIL Model 1000 Wastewater Chlorinator.*

TABLE I

SANURIL 115 Bactericidal Activity Laboratory Results Comparison with Sodium Hypochlorite

Test No.	Sample[1]	Available Chlorine, ppm[2]	pH	Avg. No. Inoculum Increments Killed[3]
Escherichia coli				
1	SANURIL 115	1.0	7.1	2
2	NaOC1	1.0	7.5	0
Salmonella typhi				
3	SANURIL 115	0.5	7.1	2
4	NaOC1	0.5	7.5	1

1. Distilled water was used in all cases. Temperature was 75° F.
2. Total available chlorine was determined by the iodometric titration method prior to testing.
3. Number of 0.05 ml. inoculum increments added at one minute intervals killed by test mixture in one minute (average of duplicate determinations).

In another test, SANURIL 115 and calcium hypochlorite were compared for effectiveness in treating sewage. A sample of effluent from an activated sludge sewage treatment plant was divided into two portions and each portion respectively, was treated with an equivalent amount of calcium hypochlorite and SANURIL 115. The results (See TABLE II) show that SANURIL 115 is comparable to calcium hypochlorite in its effectiveness against **E. coli.** However, SANURIL 115 is superior to calcium hypochlorite alone in that the synergistic blend of active ingredients provides a more rapid reduction of organisms and, a measure of control against the bacteria regrowth. Other test data is available on request.

TABLE II

Comparison of Calcium Hypochlorite and SANURIL 115 Against E. coli Present in a Treated Sewage Effluent

Sanitizer	Calcium Hypochlorite	SANURIL 115
Initial Dosage, ppm chlorine	1.7	1.7
Chlorine Residual, ppm		
at 0.5 hour	0.63	1.30
at 5 hours	0.31	0.95
at 24 hours	0.00	0.00
Initial **E. coli** conc./100 ml.	3.7×10^5	3.7×10^5
Remaining **E. coli** /100 ml.		
at 0.5 hour	2×10^2	1×10^2
at 5 hours	1×10^2	0
at 24 hours	4×10^4	0

A field test was conducted over a period of several days to evaluate SANURIL 115 as a sewage effluent disinfectant and to compare it to calcium hypochlorite. The effluent from an activated sludge sewage treatment plant was divided into two equal portions and e
was simultaneously treated with one
infecting agents.

The results in Table III show that
SANURIL 115 achieved superior bac
reductions, compared to calcium hy
and required about 20% less sanitize
job.

TABLE III

Comparison of Efficacy and Consum
SANURIL 115 and Calcium Hypoc
in Treating Sewage

	SANURIL 115
Test Period, days	16
Est. Plant Flow, gpd	3.5×10^3
Sanitizer Consumption, lbs./day	0.137
Initial **E. coli**/100 ml., range	4×10^4 – 1.14×10^5
Average **E. coli** Remaining After 0.5 Hour Detention Time[a]	625
% **E. coli** Reduction, avg.	98.3

a. Based on eight determinations.

SANURIL WASTEWATER CHLOR

SANURIL 115 Tablets were developed for use with the SANURIL WASTEWATER CHLORINATOR shown here. It was designed to be used with sewage treatment plants having capacities of up to about 100,000 gpd.

The Chlorinator consists of a dissolve
which treated sewage effluent flows, f
for applying the tablets and a weir pla
matically control sanitizer dosage. Co
of durable corrosion-resistant mate
Chlorinator is 22" long, 13" wide and
and contains four sanitizer feed tub
hold the SANURIL 115 Tablets.

CAUTION — SANURIL 115 is a strong oxidizing agent
should be taken in handling and storing the product.
warning statements and specific instructions on using
and storing SANURIL 115 are contained on the prod

Diamond Shamrock Corporation

Cleveland, Ohio 44115

Bulletin
R-SL-3